MATRIX · INVERSE

Properties of the inverse

Seven identities that mirror the rules for reciprocals — with two that do not. The order reversal and the missing sum rule are where the analogy with real numbers stops.

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IDENTITIES

CONDITIONAL 6

i Involution $(A^{-1})^{-1} = A$

CONDITIONAL

HOLDS WHEN A invertible

Inverting twice returns the original. The inverse relation is symmetric — if B is the inverse of A then A is the inverse of B, so neither matrix in the pair is privileged.

ii Inverse of a product $(AB)^{-1} = B^{-1}A^{-1}$

CONDITIONAL

HOLDS WHEN both A and B invertible

COMMON ERROR writing $(AB)^{-1} = A^{-1}B^{-1}$ — the order must reverse

The order reverses, because $(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = AA^{-1} = I$ only in that arrangement. This extends to any number of factors: $(A_1A_2\cdots A_k)^{-1} = A_k^{-1}\cdots A_2^{-1}A_1^{-1}$. The reversal is the same one the transpose performs, and both trace back to matrix multiplication not commuting.

iii Inverse of a transpose $(A^{\mathsf{T}})^{-1} = (A^{-1})^{\mathsf{T}}$

CONDITIONAL

HOLDS WHEN A invertible

The two operations commute — transpose then invert, or invert then transpose, and the result is the same. Worth contrasting with the product rule directly above: there the order matters, here it does not.

iv Scalar multiple $(cA)^{-1} = \frac{1}{c}A^{-1}$

CONDITIONAL

HOLDS WHEN A invertible and $c \neq 0$

Scalars pass through and invert. The condition $c \neq 0$ is doing real work — $0 \cdot A$ is the zero matrix, which is never invertible whatever A was.

v Powers $(A^k)^{-1} = (A^{-1})^k$

CONDITIONAL

HOLDS WHEN A invertible, k a positive integer

Follows from the product rule applied k times; the order reversal cancels out because every factor is the same matrix. This is what lets negative exponents be defined at all, with A^{-k} meaning either side of the identity.

vi Determinant of the inverse $\det(A^{-1}) = \dfrac{1}{\det(A)}$

CONDITIONAL

HOLDS WHEN $\det(A) \neq 0$

Immediate from $\det(AB) = \det(A)\det(B)$ applied to $AA^{-1} = I$. It also gives a one-line proof that a singular matrix has no inverse: there is no real number whose product with zero is one. See determinant properties.

FAILS OR UNDEFINED 1

vii Sum and difference $(A+B)^{-1} \neq A^{-1} + B^{-1}$

FAILS

FAILS WHEN in general — no rule exists

There is no product-style identity for the inverse of a sum, and $A + B$ may not even be invertible when both A and B are. The Woodbury identity handles a restricted case, but nothing does in general. This entry exists so the absence is stated rather than inferred from silence.

> **WITNESS**
>
> A = I, B = -I, both invertible
> A + B = O, which has no inverse at all

Every identity here presupposes the matrices involved are invertible, which is exactly what the equivalence above characterises. Sum and difference are absent from this list because no rule exists — that absence is the entry, not an omission.